

EV Charging Station Reservation System

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match the driver's vehicle connector type with available station chargers and reserve a 30-minute charging window.	High	Incompatible connector plugs are hidden from the driver, and two vehicles cannot book the same socket at the same time	Ensures compatible and conflict-free charging reservations.
FR-002	Functional	The system shall allow drivers to search for nearby charging stations based on location.	High	At least one compatible charging station is displayed when available.	Helps drivers find suitable charging stations.
FR-003	Functional	The system shall display available charging sockets and time slots for a selected station.	High	Only unreserved charging sockets and time slots are displayed as available.	Allows drivers to select a valid charging slot.
FR-004	Functional	The system shall allow a driver to confirm a charging station reservation.	High	A unique reservation ID is generated after successful booking.	Provides guaranteed charging access.
FR-005	Functional	The system shall calculate the charging cost using the applicable dynamic power rate	Medium	The displayed cost changes when the applicable power rate changes.	Provides transparent charging prices
NFR-001	Nonfunctional	The system shall automatically release locked sockets and apply a penalty if the driver fails to plug in within 10 minutes of slot start.	High	Testing confirms that the socket is released after 10 minutes if the driver does not plug in	Prevents charging sockets from remaining blocked unnecessarily.
NFR-002	Nonfunctional	The system shall display available charging stations and slots within 3 seconds of a search request	High	At least 95 out of 100 test searches complete within 3 seconds.	Provides a responsive user experience